TrekVerse - Trekking Management Application

Project Report

Student Details

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About:	Hi, I'm Prakruthi, a Data Science student at IIT Madras. I have a strong interest in web development and love building full-stack projects, especially with Python and Flask.

Project Details

Project Understanding:

Developed TrekVerse, a web-based Trekking Management System for efficiently managing trekking operations and bookings.

Implemented role-based access control (Admin, Staff, User) with dedicated dashboards and functionalities for each role.

Designed a complete workflow for:

- Trek creation, editing, deletion, staff assignment, and trek management by Admin
- Assigned trek management, participant tracking, slot updates, and trek status management by Staff
- Trek browsing, booking, cancellation, booking history, and profile management by Users
- Database-driven operations using Flask, SQLAlchemy, and SQLite with RESTful routes for application functionality

AI/LLM Usage

Tools Used: Google Gemini,Chatgpt

- Debugging errors and improving code correctness (<1%)
- Styling the UI using CSS and Bootstrap (8%)
- Understanding concepts and improving overall code structure

Technologies Used

Backend	Python,Flask, Flask-SQLAlchemy, Werkzeug
Frontend	HTML5, CSS3, Jinja2, Bootstrap 5
Database	SQLite
Version Control	Git,Github

DB Schema Design

Tables and Structure:

- **USER**

Relationships:

• One-to-Many with Booking
• One-to-Many with Trek (as assigned staff)

- **TREK**

Relationships:

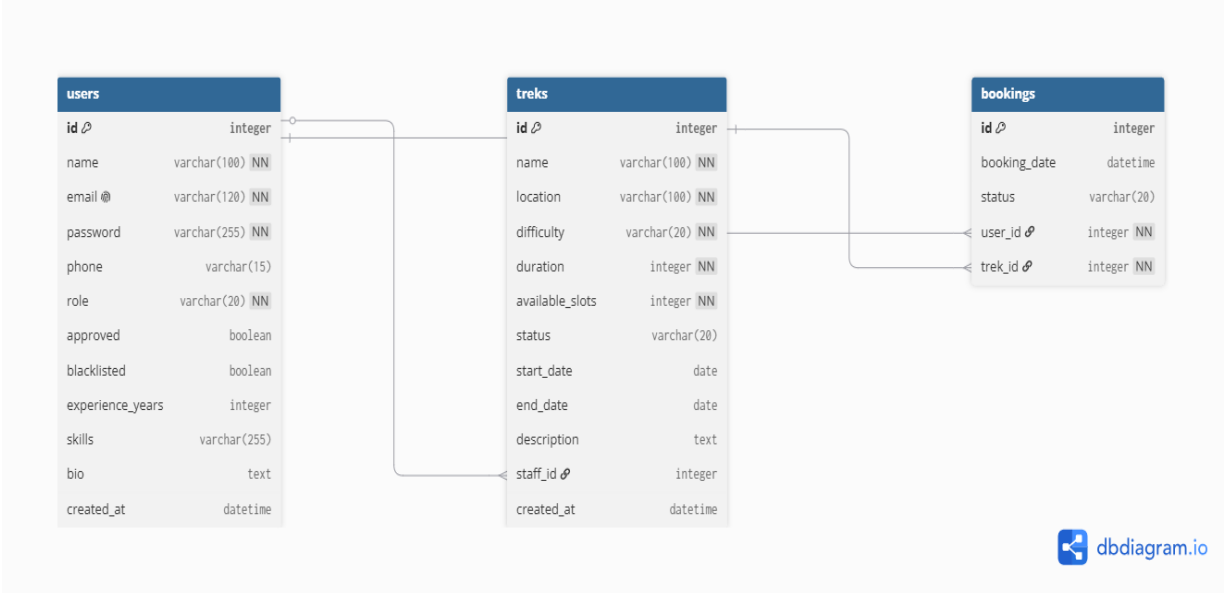
• Many-to-One with User (assigned staff)
• One-to-Many with Booking

- **BOOKING**

Relationships:

• Many-to-One with User
• Many-to-One with Trek

Schema Diagram:



Route Design

Authentication:

Method		Description
GET	/	Home page
GET,POST	/login	User login
GET, POST	/register	User registration
GET	/logout	Logout and end session

Admin:

Method	Endpoint	Description
GET	/admin/dashboard	View admin dashboard with overall statistics
GET	/admin/admin_treks	View all trekking events
GET, POST	/admin/admin_treks/add	Add a new trek
GET, POST	/admin/admin_treks/edit/<trek_id>	Edit trek information
POST	/admin/admin_treks/delete/<trek_id>	Delete a trek
GET	/admin/staff	View all staff members
GET	/admin/staff/approve/<staff_id>	Approve a staff member
GET	/admin/users	View registered users
POST	/admin/user/delete/<user_id>	Delete a user account
GET	/admin/user/toggle_blacklist/<user_id>	Blacklist or activate a user
GET	/admin/bookings	View all trek bookings
POST	/admin/booking/cancel/<booking_id>	Cancel a booking

Staff:

Method	Endpoint	Description
GET	/staff/dashboard	Staff dashboard
GET	/staff/staff_treks	View treks assigned to staff
GET, POST	/staff/staff_treks/<trek_id>	Manage an assigned trek
POST	/staff/staff_treks/<trek_id>/participants/<booking_id>/cancel	Remove a participant from a trek
GET, POST	/staff/profile	View and update staff profile

User:

Method	Endpoint	Description
GET	/user/dashboard	User dashboard
GET	/user/user_treks	Browse available treks
POST	/user/book/<trek_id>	Book a trek
GET	/user/bookings	View booked treks
POST	/user/booking/cancel/<booking_id>	Cancel a booking

Features Implemented:

User Authentication & Authorization

- Secure user registration and login

- Password hashing using Werkzeug
- Session-based authentication using Flask sessions
- Role-based access control for Admin, Staff, and User
- Predefined Admin account created programmatically
- Staff login allowed only after Admin approval

Admin Features

- Dashboard displaying total users, staff, treks, and bookings
- Add, edit, and delete trek records
- Assign staff members to treks
- View and manage all treks
- View registered users and staff
- Approve newly registered staff members
- Remove treks from the system

Staff Features

- Dedicated Staff Dashboard
- View only the treks assigned to the logged-in staff member
- View trek details including location, duration, difficulty, and available slots
- Access restricted only to approved staff members

User (Trekker) Features

- User registration and login
- User Dashboard displaying available trekking options
- Browse available/open treks
- View personal booking history
- Profile information displayed after login

Video

Video Link:
TrekVerse.mp4